

Discrete Probability Distributions

part -2

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Binomial Distributions

Binomial Experiments

A **binomial experiment** is a probability experiment that satisfies the following conditions.

1. The experiment is repeated for a fixed number of trials, where each trial is independent of other trials.
2. There are only two possible outcomes of interest for each trial. The outcomes can be classified as a success (S) or as a failure (F).
3. The probability of a success $P(S)$ is the same for each trial.
4. The random variable x counts the number of successful trials.

Notation for Binomial Experiments

Symbol

Description

n The number of times a trial is repeated.

$p = P(S)$ The probability of success in a single trial.

$q = P(F)$ The probability of failure in a single trial. ($q = 1 - p$)

x The random variable represents a count of the number of successes in n trials: $x = 0, 1, 2, 3, \dots, n$.

Binomial Experiments

Example:

Decide whether the experiment is a binomial experiment. If it is, specify the values of n , p , and q , and list the possible values of the random variable x . If it is not a binomial experiment, explain why.

- You randomly select a card from a deck of cards, and note if the card is an Ace. You then put the card back and repeat this process 8 times.

This is a binomial experiment. Each of the 8 selections represent an independent trial because the card is replaced before the next one is drawn. There are only two possible outcomes: either the card is an Ace or not.

$$n = 8 \quad p = \frac{4}{52} = \frac{1}{13} \quad q = 1 - \frac{1}{13} = \frac{12}{13} \quad x = 0, 1, 2, 3, 4, 5, 6, 7, 8$$

Binomial Experiments

Example:

Decide whether the experiment is a binomial experiment. If it is, specify the values of n , p , and q , and list the possible values of the random variable x . If it is not a binomial experiment, explain why.

- You roll a die 10 times and note the number the die lands on.

This is not a binomial experiment. While each trial (roll) is independent, there are more than two possible outcomes: 1, 2, 3, 4, 5, and 6.

Binomial Probability Formula

In a binomial experiment, the probability of exactly x successes in n trials is -

$$P(x) = {}_nC_x p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x q^{n-x}.$$

Example:

A bag contains 10 chips. 3 of the chips are red, 5 of the chips are white, and 2 of the chips are blue. Three chips are selected, with replacement. Find the probability that you select exactly one red chip.

$$p = \text{the probability of selecting a red chip} = \frac{3}{10} = 0.3$$

$$q = 1 - p = 0.7$$

$$n = 3$$

$$x = 1$$

$$\begin{aligned} P(1) &= {}_3C_1 (0.3)^1 (0.7)^2 \\ &= 3(0.3)(0.49) \\ &= 0.441 \end{aligned}$$

Binomial Probability Distribution

Example:

A bag contains 10 chips. 3 of the chips are red, 5 of the chips are white, and 2 of the chips are blue. Four chips are selected, with replacement. Create a probability distribution for the number of red chips selected.

$$p = \text{the probability of selecting a red chip} \qquad = \frac{3}{10} = 0.3$$

$$q = 1 - p = 0.7$$

$$n = 4$$

$$x = 0, 1, 2, 3, 4$$

x	$P(x)$
0	0.240
1	0.412
2	0.265
3	0.076
4	0.008

The binomial probability formula is used to find each probability.

Finding Probabilities

Example:

The following probability distribution represents the probability of selecting 0, 1, 2, 3, or 4 red chips when 4 chips are selected.

x	$P(x)$
0	0.24
1	0.412
2	0.265
3	0.076
4	0.008

a.) Find the probability of selecting no more than 3 red chips.

b.) Find the probability of selecting at least 1 red chip.

$$\begin{aligned} \text{a.) } P(\text{no more than 3}) &= P(x \leq 3) = P(0) + P(1) + P(2) + P(3) \\ &= 0.24 + 0.412 + 0.265 + 0.076 = 0.993 \end{aligned}$$

$$\text{b.) } P(\text{at least 1}) = P(x \geq 1) = 1 - P(0) = 1 - 0.24 = 0.76$$

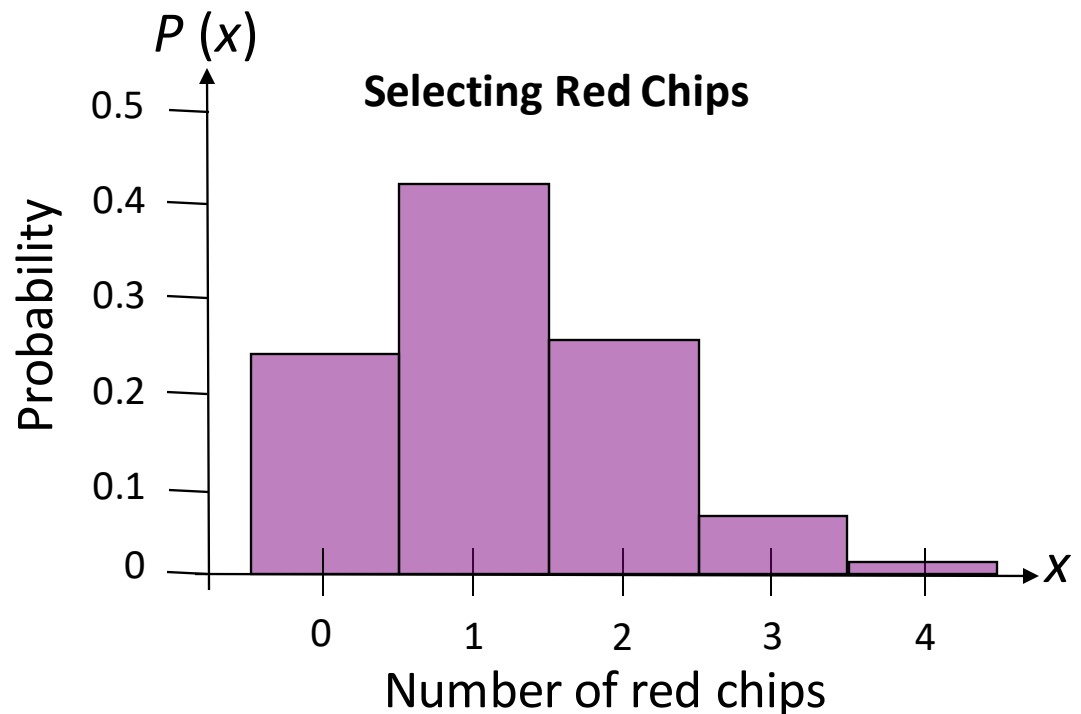
Complement

Graphing Binomial Probabilities

Example:

The following probability distribution represents the probability of selecting 0, 1, 2, 3, or 4 red chips when 4 chips are selected. Graph the distribution using a histogram.

x	$P(x)$
0	0.24
1	0.412
2	0.265
3	0.076
4	0.008



Mean, Variance and Standard Deviation

Population Parameters of a Binomial Distribution

$$\text{Mean: } \mu = np$$

$$\text{Variance: } \sigma^2 = npq$$

$$\text{Standard deviation: } \sigma = \sqrt{npq}$$

Example:

One out of 5 students at a local college say that they skip breakfast in the morning. Find the mean, variance and standard deviation if 10 students are randomly selected.

$$n = 10$$

$$p = \frac{1}{5} = 0.2$$

$$q = 0.8$$

$$\mu = np$$

$$= 10(0.2)$$

$$= 2$$

$$\sigma^2 = npq$$

$$= (10)(0.2)(0.8)$$

$$= 1.6$$

$$\sigma = \sqrt{npq}$$

$$= \sqrt{1.6}$$

$$\approx 1.3$$

More Discrete Probability Distributions

Geometric Distribution

A **geometric distribution** is a discrete probability distribution of a random variable x that satisfies the following conditions.

1. A trial is repeated until a success occurs.
2. The repeated trials are independent of each other.
3. The probability of a success p is constant for each trial.

The **probability that the first success will occur on trial x** is

$$P(x) = p(q)^{x-1}, \text{ where } q = 1 - p.$$

Geometric Distribution

Example:

A fast food chain puts a winning game piece on every fifth package of French fries. Find the probability that you will win a prize,

- a.) with your third purchase of French fries,
- b.) with your third or fourth purchase of French fries.

$$p = 0.20 \quad q = 0.80$$

$$\text{a.) } x = 3$$

$$\begin{aligned} P(3) &= (0.2)(0.8)^{3-1} \\ &= (0.2)(0.8)^2 \\ &= (0.2)(0.64) \\ &= 0.128 \end{aligned}$$

$$\text{b.) } x = 3, 4$$

$$\begin{aligned} P(3 \text{ or } 4) &= P(3) + P(4) \\ &\approx 0.128 + 0.102 \\ &\approx 0.230 \end{aligned}$$

Poisson Distribution

The **Poisson distribution** is a discrete probability distribution of a random variable x that satisfies the following conditions.

1. The experiment consists of counting the number of times an event, x , occurs in a given interval. The interval can be an interval of time, area, or volume.
2. The probability of the event occurring is the same for each interval.
3. The number of occurrences in one interval is independent of the number of occurrences in other intervals.

The probability of exactly x occurrences in an interval is -

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}$$

where $e \approx 2.71818$ and μ is the mean number of occurrences.

Poisson Distribution

Example:

The mean number of power outages in the city of Brunswick is 4 per year. Find the probability that in a given year,

- a.) there are exactly 3 outages,
- b.) there are more than 3 outages.

$$\text{a.) } \mu = 4, \quad x = 3$$

$$P(3) = \frac{4^3(2.71828)^{-4}}{3!}$$

$$\approx 0.195$$

$$\text{b.) } P(\text{more than } 3)$$

$$= 1 - P(x \leq 3)$$

$$= 1 - [P(3) + P(2) + P(1) + P(0)]$$

$$= 1 - (0.195 + 0.147 + 0.073 + 0.018)$$

$$\approx 0.567$$

THANK YOU ALL